Triangles

The big idea

Two triangles are **congruent** if they are exact copies: same size and shape. We don't need to check all six parts (three sides, three angles); certain combinations of three are enough. Congruence is the main tool for proving that lengths or angles are equal.

NOTATION

$\triangle ABC \cong \triangle PQR$ means A ↔ P, B ↔ Q, C ↔ R. Then all corresponding parts are equal: **CPCT** (corresponding parts of congruent triangles).

Congruence rules

Rule	What is equal
SAS	Two sides and the **included** angle
ASA	Two angles and the included side
AAS	Two angles and any corresponding side
SSS	All three sides
RHS	Right angle, hypotenuse and one side (right triangles)

NO SSA OR AAA RULE

Two sides and a **non-included** angle don't guarantee congruence. Three equal angles give similar, not necessarily congruent, triangles.

Properties of isosceles triangles

ISOSCELES THEOREMS

Angles opposite **equal sides** are equal. Conversely, sides opposite **equal angles** are equal.
So every angle of an equilateral triangle is 60°.

IN △ABC, AB = AC AND ∠B = 50°. FIND ∠A.

∠C = ∠B = 50° (opposite equal sides). ∠A = 180° − 100° = **80°**.

WRITING A CONGRUENCE PROOF

Given: AB = AD and AC bisects ∠BAD. Prove △ABC ≅ △ADC.
AB = AD (given), ∠BAC = ∠DAC (AC bisects ∠A), AC = AC (common). So △ABC ≅ △ADC by **SAS**, and hence BC = DC (CPCT).

PROOF LAYOUT

List three pairs of equal parts with a reason for each, name the rule, then use CPCT for what you need to prove.

Quick recap

- Congruent = identical size and shape; use CPCT afterwards.
- Rules: SAS, ASA, AAS, SSS, RHS.
- In an isosceles triangle, angles opposite equal sides are equal, and vice versa.

Practice questions

1. Which is not a congruence rule?
   (a) SAS (b) ASA (c) SSS (d) SSA
2. In $\triangle ABC$, $AB = AC$ and $\angle A = 40°$. Then $\angle B$ is
   (a) 40° (b) 70° (c) 80° (d) 100°
3. CPCT stands for
   (a) corresponding parts of congruent triangles (b) common parts of congruent triangles (c) congruent parts of corresponding triangles (d) corresponding points of congruent triangles
4. Each angle of an equilateral triangle is
   (a) 45° (b) 60° (c) 90° (d) 30°
5. Two right triangles have equal hypotenuses and one equal side. They are congruent by
   (a) SAS (b) ASA (c) RHS (d) SSS

Answers: 1. d 2. b 3. a 4. b 5. c